```
#include <WiFi.h>

#include "DHT.h"


// ------------- DHT11 SENSOR -------------

#define DHTPIN 4

#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);


// ------------- MQ GAS SENSOR PINS -------------

#define MQ5_PIN   34   // MQ-5 (LPG / Methane / Natural Gas)

#define MQ135_PIN 32   // MQ-135 (Air quality / CO2 / NH3)


// ------------- LED & BUZZER PINS -------------

#define LED_PIN     15

#define BUZZER_PIN  2


// ------------- WiFi DETAILS -------------

const char* ssid = "ESP32_AQI";

const char* password = "12345678";


WiFiServer server(80);


void setup() {

  Serial.begin(115200);

  dht.begin();


  pinMode(LED_PIN, OUTPUT);

  pinMode(BUZZER_PIN, OUTPUT);


  digitalWrite(LED_PIN, LOW);

  digitalWrite(BUZZER_PIN, LOW);
```

```
  WiFi.softAP(ssid, password);
  server.begin();

  Serial.println("IoT Air Quality Monitoring System Started...");
  Serial.print("Connect to WiFi: ");
  Serial.println(ssid);
  Serial.print("AP IP address: ");
  Serial.println(WiFi.softAPIP());
}

void loop() {
  WiFiClient client = server.available();

  // ---------- Reading MQ Gas Sensors ----------
  int mq5_raw   = analogRead(MQ5_PIN);
  int mq135_raw = analogRead(MQ135_PIN);

  float mq5_ppm   = mq5_raw   * 1000.0 / 4095.0;
  float mq135_ppm = mq135_raw * 1000.0 / 4095.0;

  // ---------- Reading Temperature & Humidity ----------
  float humidity    = dht.readHumidity();
  float temperature = dht.readTemperature();

  // ---------- GAS ALERT SYSTEM ----------
  if (mq5_ppm > 90) {      // threshold level (can be modified)
    digitalWrite(LED_PIN, HIGH);
    digitalWrite(BUZZER_PIN, HIGH);
  } else {
    digitalWrite(LED_PIN, LOW);
```

```
  digitalWrite(BUZZER_PIN, LOW);
}

// ---------- Serial Monitor Print ----------
if (!isnan(humidity) && !isnan(temperature)) {
  Serial.println("------------------------------------------------");
  Serial.println("       INDOOR AIR QUALITY MONITORING DATA       ");
  Serial.println("------------------------------------------------");
  Serial.print("MQ-5 Gas         : "); Serial.print(mq5_ppm);   Serial.println(" ppm");
  Serial.print("MQ-135 Air Quality: "); Serial.print(mq135_ppm); Serial.println(" ppm");
  Serial.print("Temperature      : "); Serial.print(temperature); Serial.println(" °C");
  Serial.print("Humidity         : "); Serial.print(humidity);   Serial.println(" %");

  if (mq135_ppm > 340)
    Serial.println("⚠ ALERT: GAS LEAK DETECTED!");

  Serial.println("air safe");
  Serial.println();
}

// ---------- MOBILE / PC DASHBOARD ----------
if (client) {
  while (client.connected() && !client.available()) delay(1);

  if (client.available()) {
    String request = client.readStringUntil('\r');
    client.readStringUntil('\n');
  }

  String html = "<!DOCTYPE html><html><head>";
  html += "<meta http-equiv='refresh' content='5'/>";
```

```
    html += "<title>Indoor Air Quality</title>";
    html += "</head><body style='font-family:Arial;text-align:center;'>";
    html += "<h1>Indoor Air Quality Monitoring</h1>";
    html += "<p><b>MQ-5 Gas:</b> " + String(mq5_ppm, 2) + " ppm</p>";
    html += "<p><b>MQ-135 Air Quality:</b> " + String(mq135_ppm, 2) + " ppm</p>";
    html += "<p><b>Temperature:</b> " + String(temperature, 2) + " °C</p>";
    html += "<p><b>Humidity:</b> " + String(humidity, 2) + " %</p>";

    if (mq135_ppm > 340)
      html += "<h2 style='color:red;'>⚠ GAS LEAK ALERT!</h2>";
    else
      html += "<h2 style='color:green;'>Air Safe ✓</h2>";

    html += "</body></html>";

    client.println("HTTP/1.1 200 OK");
    client.println("Content-Type: text/html");
    client.println("Connection: close");
    client.println();
    client.print(html);
    client.stop();
  }

  delay(2000);
}
```